Preregistration

Estimating the effect of introducing modern plant-based analogs in restaurants

Terminology

- **Dish:** A non beverage dine-in menu item, extant within the sales data, including sides and modifications to regular menu items:
  - No non food merchandise, or retail food items.
    - Examples: Commemorative cap, bottle of hot sauce, coffee.
  - No catering, take-out, or express items, unless otherwise indistinguishable from dine-in sales within the data.
    - Examples: Order of 36 sub sandwiches, unlisted apple from a basket.
  - No drinks, alcoholic or nonalcoholic. However, we count smoothies as dishes rather than as beverages.
  - No itemized sides and condiments or sales below $2.00 per unit.
- **Menu**: The set of all dishes matched as closely as possible to ground-truth menus. Ground-truth menus are found by cross-referencing online menus that reflect menus found within in-person dining. In cases where cross-reference is difficult, only items with more than 10 sales within the data will be considered.
- **Animal-based food (ABF):** Any animal product, such as all meats (red and white), seafood (fish and shellfish), dairy, and eggs.
- **Modern plant-based analog (MPBA):** A modern vegan product designed to *closely* emulate a specific animal-based food *by design* or *out of the package*.
  - Examples: Impossible burgers, JUST egg, Daiya cheese, Lightlife crumbles, Black Sheep lamb.
  - Non-examples: Tofu, tempeh, black bean burgers, falafels, unbranded seitan, soy curls.
- **MPBA-containing dish**: A dish *containing* a MPBA. The dish need not be entirely vegan or even vegetarian.
- **MPBA-modifiable dish**: Due to both low counts of MPBA-containing dishes and definitional ambiguity of determining whether a given dish contains an MPBA, we also consider whether dishes can have an MPBA as a canonical option. A canonical option is

one that is not necessarily the default, but is at least officially advertised by the restaurant as an option.

- **Menu change (MPBA or vegan):** A change in menu offering that affects the number of dishes offered of a particular type, such as MPBA or vegan.
    - Examples: Adding a new MPBA-containing item to the menu changes the count and proportion of MPBA-containing dishes on the menu. Adding a meat dish to the menu affects the proportion of vegan dishes by negation.
- **Vegan dish**: Any dish containing no animal products, except for in trace amounts. Determining whether a dish is vegan is done by referencing its menu description. If menu descriptions are unavailable, the default label will be based on the prominence and the standard ingredients commonly used in that dish. Enumerated, we use the:
    - 1. Menu description
    - 2. Prominence in dish name and description
    - 3. Common ingredients for the dish

  Dishes with trace amounts of collagen powder, whey, gelatin, and honey will be considered vegan. However, if there is a comparable vegan ingredient for an animal ingredient, any dish with the nonvegan ingredient will always be considered as animal-based in its entirety.
    - Examples: Acai smoothie with soy milk and honey. Mushroom aioli sandwich when the aioli is found to be dairy-free on the menu. Dishes with unlisted dairy aioli. Salads with a vinaigrette that has dairy.
    - Nonexamples: Salads with ranch dressing. Mushroom aioli sandwich without a listing since i) aioli is made prominent and ii) aiolis often have dairy. A smoothie with collagen powder is considered a meat dish when there is an explicit vegan collagen powder on the menu, and otherwise considered vegan.
- **Non-vegan dish:** Any dish not meeting the criteria for a vegan dish.
- **Vegetarian dish:** Any dish containing no meat, i.e., red meat, poultry, seafood (thus this category is a superset of the vegan dish category).
- **Meat dish**: Any dish not meeting the criteria for a vegetarian dish.
- **Targeted:** For a given MPBA, a targeted non-vegan dish is one that is a direct animal-based analog to it
    - Examples: Impossible burgers for red meat burgers, Violife cheese slices for dairy cheese slices, pulled jackfruit for pulled pork, vegan chicken burger for chicken burger, Impossible sausage patty for sausage patty.
    - Nonexamples: Impossible burger for hotdog, Impossible sausage patty for red meat burger, Impossible chicken nuggets for red meat burgers, Morningstar breaded chicken patty for red meat burgers.

Research Overview

Primary research questions

(Q1) In real restaurants, to what extent does introducing plant-based analogs (PBA) that closely emulate animal-based foods (ABF) reduce consumption of ABF?
(Q2) To what extent does the proportion of such items on a menu reduce ABF consumption?
(Q3) Do PBAs primarily displace the specific ABF they are intended to emulate or do they generally displace other ABFs?
Secondary questions:
(Q4) Are displacement effects, if existing, moderated by restaurant characteristics (e.g., baseline proportion of ABF on the menu, cuisine type) or characteristics of the PBA itself (e.g., type of ABF being emulated; similarity to an existing ABF menu item; absolute price and price relative to the rest of the menu)?

Data collection

We have rich, daily data on the number of sales of every menu item, their prices, and customer IDs. For the majority of restaurants within the dataset, the data covers at least 2-6 months before and after the PBA introduction.

Statistical Analyses

There will be a total of six sets of analyses and additional exploratory analyses.

Of these six, there will be four sets of analyses that estimate the net causal effect of MPBA-related exposures on ABF orders, which includes any effects of the MPBAs on the customer base (e.g., attracting veg*n customers who would not otherwise have gone to the restaurant). There will additionally be two sets of analyses that estimate the average individual-level causal effect of the same exposures on the same outcomes.

The focus of these six analyses can be equivalently recast as two analyses aimed at measuring the net causal effects of the overall menu presence of MPBAs, two analyses aimed at measuring the net causal effects of the introduction of specific MPBAs through interrupted time series (ITS), and two analyses aimed at measuring customer-specific average causal effects through ITS.

Listed out, they are:

- (A1&A2) Analysis of the effects of 1. overall MPBA and 2. animal-specific MPBA menu presence on general ABF dish sales
- (A3&A4) Interrupted time-series analysis of MPBA introductions on 3. general and 4. targeted ABF sales
- (A5&A6) Interrupted time-series analysis of MPBA introductions on 5. general and 6. targeted ABF sales for a fixed-customer-base

Data: Preparation

Data inclusion and model specification, to be discussed in the following sections, were decided upon through extensive data processing and exploratory analysis. Restaurant sales data is large-scale and often suffers data issues including timezone discrepancies, idiosyncratic naming schemes, and highly detailed item modifications. Many small restaurants in particular lack systems for recording dish sales clearly and coherently. To manage such issues, we cleaned the data extensively and leveraged external resources for validation, including but not limited to dish renaming, non-food removal, and the manual labeling of thousands of dish sales with varying modifications.

Throughout, exploratory data analysis was an important tool for assessing the ongoing refinement of the data and potential model specifications. Such exploratory analysis helped us iterate upon initial research questions and define models well-calibrated to the data. In fitting models, we deliberately held out any information pertaining to exposures of interest, including the introduction of new MPBAs and the menu quantities of MPBAs. Given the complexity of the sales data, it was crucial to evaluate model fit in some way before committing to a finalized set of analyses. Fit models were chosen from increasingly expressive model classes as simple models failed to capture important aspects of data trends. As more features were included within the models, we improved fits while avoiding data leakage and related issues.

Without thorough data cleaning / exploratory analysis and the insight that comes from this familiarity, it would not have been possible to formulate empirically grounded, research-relevant models.

Data: Inclusion & Exclusion Criteria

Data Quality

The dataset includes twenty restaurants. However, data quality between restaurants is variable, and some have clearly higher quality than others. We will therefore split each of the six analyses into two tiers of subanalyses: Tier One and Tier Two. This amounts to twelve subanalyses. To define the tiers, we will separate the restaurants into two sets, a first set (Set One) for inclusion in Tier One, and a second set (Set Two), to be *combined with* Set One for inclusion in Tier Two. In simple terms, Tier Two subanalyses includes all the restaurants, those with both higher and lower quality data while Tier One subanalyses include only restaurants with high quality data.

Inclusion within Set One will be based on a stringent quantitative metric developed from exploratory analysis with the datasets: **90% or higher daily coverage**. Daily restaurant coverage is defined as having at least one sale every weekday. While some restaurants have nonstandard weekly schedules that would seem to artificially lower this metric, the daily coverage metric empirically aligns with other data integrity measures, such as week coverage, half-day coverage, gap-between-sales counts, and other general data quality statistics.

The following seven restaurants that meet the 90% threshold will be defined as Set One:
1. VSOALUACMZ, 2. SRQS8F7JWA9MZ, 3. 2HRX9P6HKXA8V, 4. JHDN7CF1C03X5, 5. L69HYJ4Y3TR91, 6. ED5J990H5VAZT, 7. W8T41JZK0ZMEP

The remaining thirteen will be defined as Set Two by failing to meet that threshold. In addition to this threshold, other issues listed in the following subsections determine further modifications to Set One and Set Two inclusion depending on analysis-specific requirements.

A1 & A2: Data Screening for Menu Presence Analyses

Restaurants must have enough changes to their menu to be analyzable by menu presence, as follows:

Restaurants will be excluded from menu presence analyses (A1 & A2) entirely when:

- There are less than **one MPBA menu change** or less than **three vegan menu changes** during the timeline of the dataset.

One restaurant met this exclusion criterion from analysis thus far.

Determining changes in menu presence requires knowledge of a ground-truth menu at every point in time. Reconstructing such a menu for points in time multiple years prior to present can be difficult, depending on the availability of online information. Furthermore, menu items within the data do not match canonical menu items within the restaurant due to idiosyncratic entry by staff. Even without idiosyncratic entry, certain menu items may or may not represent something that constitutes a dish (as defined in the Terminology). Thus, properly reconstructing the menu requires both the consolidation of menu items within the data and the matching of those items with official menus past and present. In the event that consolidation and matching proves infeasible or unreasonable for a given restaurant, the restaurant will be excluded from A1 & A2. If consolidation and matching seems to lower the data quality for a restaurant within Set One, that restaurant will be relegated to Set Two for the menu presence analyses.

Furthermore, if a restaurant is revealed to have fewer MPBA-containing menu changes than expected, it will either be relegated to Set Two or excluded entirely from menu presence analysis.

A3 – A6: Data Screening for ITS Analyses

ITS models will attempt to capture every new MPBA-containing dish that occurs at a separate time. If two MPBA dishes are introduced at the same time, they will be counted as one "shock". If they are introduced at different times, they will be counted as two separate shocks within the same restaurant. However, confounders other than simultaneous MPBA introductions are an issue.

Restaurant-MPBA combinations will be excluded from all ITS analyses (A3 – A6) when:

- There is less than **two months of data before and after** introducing that MPBA,
- The introduced PBA is **not considered modern**.

Two restaurants met these exclusion criteria from analysis thus far.

Restaurant-MPBA combinations will be relegated from Set One to Set Two for inclusion within the Secondary analyses when:

- There are more than **two or more food items** introduced contemporaneously (i.e., within **one month before or after)** introducing a MPBA and in which no other major relevant menu changes were made (where relevant means it could plausibly upweight or downweight the MPBA sales).

One restaurant met this exclusion criterion from analysis thus far.

Menu reconstruction is not important for the ITS analyses. However, what's important for the ITS analyses is the timing of the exposures (the introduction of the MPBAs). Determining the introduction dates is difficult and requires external information in a similar way as the menu presence analyses. In the event that there is evidence that entirely contradicts an introduction date, that restaurant-MPBA combination will be excluded from the ITS analyses. Finally, in the event that certain dishes have unclear start dates, conflicting start dates according to other information, start dates with extremely sparse data (due to data quality, not low sales), or start dates that seem intermixed with other dishes, these dishes will be excluded from the ITS analyses. However, being MPBA items, their influence will be addressed in the menu presence analyses.

If new MPBA dishes are found after ground-truth menu consolidation such that these dishes were introduced during the time of the dataset *and* these dishes don't meet any of the above exclusion criteria, then those dishes will be added to the ITS analyses.

A5 – A6: Data Screening for Fixed-Customer-Base ITS Analyses

Restaurants that do not have sufficient gender data on customers will be excluded, as follows:

- As long as 5 percent of the customers that have purchase data before and after the introduction date of MPBAs have gender data (and as long as this totals to 50 or more), those restaurants will be included in the analysis.

Resulting Restaurant Inclusion List

Based on these inclusion and exclusion criteria, restaurants with the following anonymous identifiers will be included in these analyses, respectively:

Menu Presence Analyses A1 & A2

Menu Presence Analyses, Set One

1. SRQS8F7JWA9MZ
2. 2HRX9P6HKXA8V
3. JHDN7CF1C03X5

4. L69HYJ4Y3TR91
5. ED5J990H5VAZT
6. W8T41JZK0ZMEP

VSOALUACMZ is excluded due to no menu changes.

Menu Presence Analyses, Set Two

7. EMBVNVD207CC6
8. C0BE4NDSW26QN
9. V3Q26BHF3SE2H
10. LBZEEFSBJNB3Z
11. SAFK7ND1HR6XS
12. S8MT0YGD2KTN9
13. LFZFT3VASXPED
14. 1SQPTEGYPH0GA
15. 9XKJD8DQTH559
16. LQ5EH4BKGV61T
17. 78AY09MVJVTYE
18. 75WYSXR9QBK5M
19. CB2KHY1C2G9PT

ITS Analyses A3 & A4

Dishes on the same line are introduced at the same time.

ITS Analyses, Set One

1. VSOALUACMZ
    a. Black Sheep Lamb Sandwich, Black Sheep Lamp Salad
2. SRQS8F7JWA9MZ
    a. Impossible Breakfast Sandwich, Impossible Patty Melt
    b. Beyond Burger
3. 2HRX9P6HKXA8V
    a. Beyond Sausage
4. JHDN7CF1C03X5
    a. Beyond Burger, Vegan Cheese
    b. Beyond Sausage
5. L69HYJ4Y3TR91
    a. Impossible Sausage Patty
6. ED5J990H5VAZT
    a. Vegan Bacon, Make It Vegan

W8T41JZK0ZMEP is relegated to Set Two due to confounding menu introductions.

ITS Analyses, Set Two

6. W8T41JZK0ZMEP
    a. Vegan Breakfast Sandwich
    b. Vegan Cupcake
    c. Vegan Cheesecake
    d. Raspberry Lemon No-Bake Bar
    e. Vegan Cake Slice
7. EMBVNVD207CC6
    a. Vegan Cheese Pizza
8. C0BE4NDSW26QN
    a. Beyond Burger
    b. Impossible Burger
9. V3Q26BHF3SE2H
    a. Sausage Egg & Cheese, Chipotle Sausage Egg and Cheese, Southwest Panini
    b. Wings,
    c. South West Sausage Egg and Cheese
    d. Vegan SEC
10. LBZEEFSBJNB3Z
    a. Wake And Fake (Vegan Cheese and Vegan Sausage)
11. SAFK7ND1HR6XS
    a. Jackfruit Tacos
12. S8MT0YGD2KTN9
    a. Vurger
13. 1SQPTEGYPH0GA
    a. Impossible Meatballs
14. 9XKJD8DQTH559
    a. Impossible Burger
    b. Vegan Cheese
15. LQ5EH4BKGV61T
    a. Beyond Burger
16. 78AY09MVJVTYE
    a. Veggie Sausage

LFZFT3VASXPED, 75WYSXR9QBK5M and CB2KHY1C2G9PT are excluded due to no data before the introduction of a PBA, a lack of a *modern* PBA, and no data before the introduction of a PBA, respectively.

ITS Analyses A5 & A6

ITS Analyses, Set One

1. SRQS8F7JWA9MZ
    a. Impossible Breakfast Sandwich, Impossible Patty Melt

    b. Beyond Burger
2. 2HRX9P6HKXA8V
    a. Beyond Sausage
3. L69HYJ4Y3TR91
    a. Impossible Sausage Patty
4. ED5J990H5VAZT
    a. Vegan Bacon, Make It Vegan
5. W8T41JZK0ZMEP
    a. Vegan Breakfast Sandwich
    b. Vegan Cupcake
    c. Vegan Cheesecake
    d. Raspberry Lemon No-Bake Bar
    e. Vegan Cake Slice

ITS Analyses, Set Two

6. EMBVNVD207CC6
    a. Vegan Cheese Pizza
7. C0BE4NDSW26QN
    a. Beyond Burger
    b. Impossible Burger
8. V3Q26BHF3SE2H
    a. Sausage Egg & Cheese, Chipotle Sausage Egg and Cheese, Southwest Panini
    b. Wings,
    c. South West Sausage Egg and Cheese
    d. Vegan SEC
9. LBZEEFSBJNB3Z
    a. Wake And Fake (Vegan Cheese and Vegan Sausage)
10. SAFK7ND1HR6XS
    a. Jackfruit Tacos
11. S8MT0YGD2KTN9
    a. Vurger
12. 1SQPTEGYPH0GA
    a. Impossible Meatballs
13. 9XKJD8DQTH559
    a. Impossible Burger
    b. Vegan Cheese
14. LQ5EH4BKGV61T
    a. Beyond Burger
15. 78AY09MVJVTYE
    a. Veggie Sausage

In addition to exclusions for A3 and A4, VSOALUACMZ will be excluded from the customer ITS because it does not have sales entries containing customer information. JHDN7CF1C03X5 will further be excluded because it does not have any gender data for customers that had purchases both before and after MPBA introductions.

Modeling

Modeling Overview

To recapitulate, there are a total of six analyses (A1 – A6), and twelve subanalyses due to repeating each analysis for Tier One (T1) and Tier 2 (T2). Within each subanalysis, we will fit models for both primary and secondary outcomes. Primary outcomes are those that pertain most directly to whether MPBA and other menu-based exposure change animal product sales. Effects of each exposure of interest will be estimated within different models. Therefore, for every outcome and every exposure, we will fit a multilevel model, denoted Global, that includes all restaurants. This Global model will estimate the total, restaurant-pooled effect, as well as restaurant-specific random effects.

As a robustness check, we will also fit single-restaurant models, each denoted Local, for a subset of the analyses and outcomes. For that subset, we will compare multilevel restaurant-specific effects with effects estimated by the Local models. In the case that the Global model (for a given outcome) fails to converge, we will use a more simplified covariate set (specifically, the covariate set C_t defined in Analysis Set 1 below, would be more simplified). In the case that the simple model fails to converge, we will rely on single-restaurant Local models. In that case, we will estimate the global result from all the Local models by conducting a meta-analysis on the restaurant-level estimated effects and aggregating them.

For the ITS analyses (A3 – A6) in particular, there will be an additional level in the modeling hierarchy to capture there being multiple MPBA introductions in individual restaurants. The coefficients for these dish introductions will be partially pooled not only across restaurants as in the other Global models, but also within restaurants. If a model in an ITS analysis fails to converge, then we will attempt to fit the Global model with only partial pooling across restaurants and not within. In that case, we will only model one dish introduction for each restaurant, choosing the first dish listed for it within the Resulting Restaurant Inclusion Lists.

For all analyses, we will evaluate and present the results according to tiered evidentiary thresholds. Within each of the twelve subanalyses, we will show p-values without multiplicity correction and will indicate significance at Bonferroni-corrected thresholds of two different stringency levels. The first Bonferroni-corrected alpha will be determined within each analysis separately, so the denominator will be the number of primary outcomes. For example, if there are 18 coefficients of primary interest, they would be corrected at alpha=0.00278 (0.05/18). The second Bonferroni-corrected alpha will count primary outcomes across all twelve subanalyses. Each resulting coefficient estimate will be annotated (e.g., in tables) according to the threshold(s) it surpasses.

The models are specified in the Stan programming language (through R) and sampled using Stan's No U-Turn Sampler (NUTS), an MCMC algorithm. We will run at least 3 parallel chains per model, with at least 1000 iterations. MCMC tree-depth, jump adaptivity, and exact initializations for warm-up and sampling iterations will depend on convergence success.

As an algorithm that reflects the entire process, we have:

- For every analysis A1–A6:
- Split the data into the Tier One and Tier Two (i.e., Set One and Set One + Set Two)
- Then split the dataset into primary and secondary outcomes depending on the analysis
- For every outcome within the analysis set
- Fit the model to training data
- Generate rolling single-step forecasts on out-of-sample testing data using MCMC

A schematic for the different levels of the analysis is on the following page.

The total number of models depends on the analysis. For each of the six main analyses, there are a different number of exposures of interest and outcomes. The total number of models for each analysis is listed in each subsection below.

Schematic

- A1 (Analysis - Distinct Aim and Model Formulation)
  - Tier 1 (Subanalysis - Subset of Restaurants Based on Data Quality)
    - Outcome 1 (Outcome-Level Model Set)
      - Exposure 1 (One Model per Exposure-Outcome Pair)
      - ...
      - Exposure k
    - ...
    - Outcome n
      - Exposure 1
      - ...
      - Exposure k
  - Tier 2 (Subanalysis - Full Set of Restaurants)
    - Outcome 1
      - Exposure 1
      - ...
      - Exposure k
    - ...
    - Outcome n
      - Exposure 1
      - ...
      - Exposure k
- ...
- A6
  - Tier 1
    - Outcome 1
      - Exposure 1
      - ...
      - Exposure k
    - ...
    - Outcome n
      - Exposure 1
      - ...
      - Exposure k
  - Tier 2
    - Outcome 1
      - Exposure 1
      - ...
      - Exposure k
    - ...
    - Outcome n
      - Exposure 1
      - ...
      - Exposure k

Base Model & Estimation

For each subanalysis, we will use the same base model, regressing the outcomes of interest on the covariates (C_t) and a single exposure of interest. The single exposure of interest will change depending on the analysis. For all models, time points, *t*, will be defined at the day level. The details are as follow:

We will employ a multilevel INGARCH(p,q) model, having the outcomes distributed according to the negative binomial distribution, and additionally log parametrized. INGARCH models specialize in modeling time series count data (non-negative integer-valued variables). As time series models, INGARCHs model autoregression in the outcome and in the latent mean/intensity parameter. Modeling the latent intensity addresses unmeasured autocorrelation in the errors, akin to the way ARMA models model moving averages. Heteroskedasticity is naturally addressed through the implicit variance modeling within the design. We use the negative binomial variant of INGARCH (over the Poisson variant) to better fit overdispersion in the outcome, apparent within this sales data and common to sales data in general. This model structure was chosen without knowledge of or reference to the introduced MPBA exposures.

The multilevel / mixed effects structure will include all restaurants that meet the inclusion criteria, and will include random intercepts and slopes by restaurant for a subset of the predictors, lagged outcomes, and lagged latent intensities. For the ITS analyses, each introduced MPBA will have its own effect modeled separately and partially pooled within the restaurant, so there will be two levels to the intercepts and slopes, one level by MPBA within the restaurant and one by restaurant.

While we initially explored model validation through rolling K-fold cross-validation and a custom space-filling grid search, we have elected to circumvent post-selection and mitigate its accompanying bias by fitting an extensive superset model that contains all predictors and all highly plausible time-dependent lags. In this data, the plausible time-dependent lags are 1-7 days prior and 1-8 weeks prior. We avoid formally selecting predictors or lags, and we attempt to induce sparsity in the model through multilevel Bayesian LASSO (or equivalently stated, by specifying a Laplace prior on the multilevel location parameters for both predictor and lags). In practice, this amounts to two levels of regularization, first towards the mean across restaurants, and second towards zero through the Laplace. If a Laplace prior on the location parameters leads to nonconvergence, a Gaussian prior can instead be used.

Global

The Global model containing all restaurants includes restaurant-level clustering and can be presented as follows:

Observation Model

For each restaurant $r = 1, \dots, R$, over time $t = 1, \dots, T_r$:

$$y_{t,r} \sim \text{NegBinom}_2(\lambda_{t,r}, \phi_r)$$

With the additive structure being restaurant-specific:

Latent Process Model

$$\log(\lambda_{t,r}) = \sum_{j \in \mathcal{J}_{\text{fixed}}} \beta_j C_{t,r}^{(j)} + \sum_{j \in \mathcal{J}_{\text{rand}}} \beta_{j,r} C_{t,r}^{(j)} + \sum_{e=1}^{E_r} \gamma_{e,r} X_{t,r}^{(e)} + \sum_{k=1}^{p'} \alpha_{k,r} \log(y_{t-\ell_{\alpha,k},r} + 1) + \sum_{l=1}^{q'} \delta_{l,r} \nu_{t-\ell_{\delta,l},r}$$

Here, $\mathbf{C}_{t,r}$ is a vector of J non-exposure covariates for restaurant r at time t, with coefficients $\boldsymbol{\beta}_r$. The covariate coefficients may be fixed across restaurants or vary by restaurant, depending on the index set $j \in \mathcal{J}_{\text{fixed}} \cup \mathcal{J}_{\text{rand}}$.

The matrix $\mathbf{X}_{t,r} = (X_{t,r}^{(1)}, \dots, X_{t,r}^{(E_r)})$ represents E_r exposure-specific covariates for restaurant r, each with a coefficient $\gamma_{e,r}$ drawn from a nested hierarchical prior.

The terms p' and q' denote the number of autoregressive lags applied to past log-counts and latent log-rate values, respectively. The lags $\ell_{\alpha,k}$ and $\ell_{\delta,l}$ specify which time points are used for each term.

Viewed as a probabilistic model, its posterior is:

Probabilistic Model

$$P(\text{all parameters} \mid \mathbf{Y}, \mathbf{C}, \mathbf{X}) \propto \left[\prod_{r=1}^{R} \prod_{t=1}^{T_r} P(y_{t,r} \mid \lambda_{t,r}, \phi_r) \right] \times P(\text{priors})$$

The overdispersion parameter be modeled as:

- **Overdispersion ϕ_r:**
  Each ϕ_r models negative binomial overdispersion within each restaurant:

$$
\begin{aligned}
\phi_r &\sim \text{LogNormal}(\mu_{\log\phi}, \sigma_{\log\phi}) \\
\mu_{\log\phi} &\sim \text{Normal}(0, s_{\mu\log\phi}) \\
\sigma_{\log\phi} &\sim \text{Exponential}(s_{\sigma\log\phi})
\end{aligned}
$$

The coefficients for some covariates will vary across restaurants, while others will be modeled as fixed across restaurants. They will respectively be modeled as:

- **Covariate Coefficients $\boldsymbol{\beta}_r = (\beta_{1,r}, \ldots, \beta_{J,r})^T$:**
  Each coefficient $\beta_{j,r}$ is modeled based on whether it is fixed or varies across restaurants:
  - If $\beta_{j,r}$ is a coefficient fixed across all restaurants (i.e., $\beta_{j,r} = \beta_j$):

$$
\begin{aligned}
\beta_j &= \mu_{\beta_j} \\
\mu_{\beta_j} &\sim \text{Normal}(0, s_{\mu\beta})
\end{aligned}
$$

  - If $\beta_{j,r}$ is an intercept or a restaurant-varying coefficient:

$$
\begin{aligned}
\beta_{j,r} &\sim \text{Normal}(\mu_{\beta_j}, \sigma_{\beta_j}) \\
\mu_{\beta_j} &\sim \text{Laplace}(0, s_{\mu\beta}) \\
\sigma_{\beta_j} &\sim \text{Exponential}(s_{\sigma\beta})
\end{aligned}
$$

The exposures of interest will be modeled the same as the restaurant-varying coefficients for the non-ITS models, and will be modeled for the ITS models as:

- **Exposure Coefficients $\boldsymbol{\gamma}_r = (\gamma_{1,r}, \ldots, \gamma_{E_r,r})^T$:**
  Let $\mathbf{X}_{t,r}$ be the E_r-dimensional vector of exposure covariates for restaurant r at time t, with each column corresponding to one exposure. Each coefficient $\gamma_{e,r}$ is partially pooled within and across restaurants:
  - For the exposure coefficient:

$$
\begin{aligned}
\gamma_{e,r} &\sim \text{Normal}(\eta_r, \sigma_{\gamma,\text{within}}) \\
\eta_r &\sim \text{Normal}(\mu_\gamma, \sigma_{\gamma,\text{between}}) \\
\mu_\gamma &\sim \text{Normal}(0, s_{\mu\gamma}) \\
\sigma_{\gamma,\text{within}} &\sim \text{Exponential}(s_{\sigma\gamma,\text{within}}) \\
\sigma_{\gamma,\text{between}} &\sim \text{Exponential}(s_{\sigma\gamma,\text{between}})
\end{aligned}
$$

The autoregressive coefficients for the outcome are below. The tanh transformation properly compresses each alpha to lie between -1 and 1, which is required by the log parameterization (the factor of 2 prevents excessive compression).

- **Autoregressive Coefficients** $\boldsymbol{\alpha}_r = (\alpha_{1,r}, \ldots, \alpha_{p',r})^T$**:**
  Coefficients $\alpha_{k,r} \in (-1, 1)$ because of the tanh-transformation:
  - If $\alpha_{k,r}$ is fixed across all restaurants (i.e., $\alpha_{k,r} = \alpha_k$):

$$\alpha_k = \tanh(\mu_{\alpha_k}/2), \quad \text{where } \mu_{\alpha_k} \sim \text{Normal}(0, s_{\mu\alpha})$$

  - If $\alpha_{k,r}$ is restaurant-varying:

$$\begin{aligned}
\alpha_{k,r} &= \tanh(\tilde{\alpha}_{k,r}/2), \quad \text{where } \tilde{\alpha}_{k,r} \sim \text{Normal}(\mu_{\alpha_k}, \sigma_{\alpha_k}) \\
\mu_{\alpha_k} &\sim \text{Laplace}(0, s_{\mu\alpha}) \\
\sigma_{\alpha_k} &\sim \text{Exponential}(s_{\sigma\alpha})
\end{aligned}$$

The autoregressive coefficients for the latent intensity are:

- **Autoregressive Coefficients** $\boldsymbol{\delta}_r = (\delta_{1,r}, \ldots, \delta_{q',r})^T$**:**
  Coefficients $\delta_{l,r} \in (-1, 1)$ because of the tanh-transformation:
  - If $\delta_{l,r}$ is fixed across all restaurants (i.e., $\delta_{l,r} = \delta_l$):

$$\delta_l = \tanh(\mu_{\delta_l}/2), \quad \text{where } \mu_{\delta_l} \sim \text{Laplace}(0, s_{\mu\delta})$$

  - If $\delta_{l,r}$ is restaurant-varying:

$$\begin{aligned}
\delta_{l,r} &= \tanh(\tilde{\delta}_{l,r}/2), \quad \text{where } \tilde{\delta}_{l,r} \sim \text{Normal}(\mu_{\delta_l}, \sigma_{\delta_l}) \\
\mu_{\delta_l} &\sim \text{Laplace}(0, s_{\mu\delta}) \\
\sigma_{\delta_l} &\sim \text{Exponential}(s_{\sigma\delta})
\end{aligned}$$

Analysis set 1 (A1): Effects of MPBAs' overall menu presence on general ABF dish sales

Variables

Primary outcomes (Y_t):
1. Absolute non-vegan dish sales (count with no offset)
2. Absolute meat dish sales (count with no offset)
3. Absolute sales of chicken and fish combined (count with no offset)

Secondary outcomes (Y_t):

4. Absolute vegan dish sales (count with no offset)
5. Absolute vegetarian dish sales (count with no offset)
6. Total dish sales (count with no offset) - as a sensitivity analysis to look at changes in customer base

Exposures of interest (X_t):
1. Proportion of vegan menu options
2. Proportion of vegetarian menu options
3. Proportion of MPBA-modifiable menu options
4. Number of vegan menu options
5. Number of vegetarian menu options
6. Number of MPBA-modifiable menu options

Covariates (C_t):
1. Average price of each vegan dish sold over the past 7 days, *inflation-adjusted* (continuous)
2. Average price of each non-vegan vegetarian dish sold over past 7 days, *inflation-adjusted* (continuous)
3. Average price of each meat-containing dish sold over the past 7 days, *inflation-adjusted* (continuous)
4. Inflation (continuous)
5. Local temperature (continuous)
6. Local precipitation (continuous)
7. Day of week (categorical)
8. Weekend (categorical)
9. Holidays (categorical)
10. Month (categorical)
11. Season (categorical)
12. Year (categorical)
13. Date (continuous); this accounts for secular trends and is especially important in the ITS analysis

In the event that certain covariates are highly collinear (i.e., completely preventing convergence - other collinearity is fine) for a model (e.g., inflation and year are collinear, or year and date are), variables will be removed. In the event that certain covariates can not be reasonably used due to a lack of existing levels (e.g., a restaurant not having multiple years of data), that variable will be removed.

Note regarding the price covariates: We are not controlling for the prices of dishes on the menu because we don't have the actual menu at each point in time; this approach effectively controls for the average price of the dishes actually sold, weighted by the quantity sold.

Note regarding mixed MPBAs: If MPBA is mixed (e.g., Impossible Burger with animal cheese) and Y is vegetarian dishes, the MPBA is counted in both exposure and outcome.

Note regarding definitions: We won't include all exposures in the same model because their definitions overlap (e.g., MPBA dishes and vegan dishes).

Total number of coefficients of interest per model and number of models

Coefficients of primary interest with Tier 1 (3 primary outcomes * 6 exposures) = 18 models, 18 coefficients

Total coefficients (primary or secondary):
Tier 1: (18 primary outcomes + 18 secondary outcomes) * (1 all-restaurants model) = 36 models
Tier 2: (18 primary outcomes + 18 secondary outcomes) * (1 all-restaurants model) = 36 models

Total: 72 models, 72 coefficients

Analysis set 2 (A2): Effects of animal-specific MPBA menu presence on the targeted ABF

Variables

Outcomes (Y_t):

1. Number of sales including animal product A_i (defined below)

Exposure (X_t):

1. Presence of at least one MPBA that emulates animal product A_i
2. Number of MPBA options that emulate animal product A_i

Covariates (C_t):

1. Average price of all A_i-emulating MPBAs on the restaurant's menu
   Covariates listed for Analysis 1

The following animal product categories were chosen to best fit the dataset and its MPBAs.

<u>Animal product categories (AP_categories):</u>

1. Lamb chunked meat (lamb)
2. Red meat burger (beef/pork)
3. Sausage (beef/pork)
4. Meatballs (beef/pork)
5. Bacon (pork)
6. Ground meat (beef/pork)
7. Breakfast sandwich sausage (beef/pork)
8. Chunked meat (beef/pork)
9. Pulled pork (pork)
10. Unfried chicken
11. Fried chicken/fried wings (chicken/poultry)
12. Savory food-based dairy (cheese/cream cheese)
    a. E.g., pizza, bagels
13. Sweet food-based dairy (plant milk/chocolate/plant butter)
    a. Eg., cupcakes, brownies
14. Egg

There were no analogs for seafood (fish/shellfish) in the dataset. A new category will be added if a new MPBA does not clearly fall within any. These twelve categories can be condensed down into six broader classes to have a total of six multilevel models for each Tier 1 and Tier 2. These classes were chosen based on product similarity, though empirical research into the food groups that best represent substitutability is somewhat limited.

<u>Animal product classes (A_i):</u>

1. Whole meat - *like muscle*
    a. Lamb chunked meat (lamb)
    b. Chunked meat (beef/pork)
    c. Pulled pork (pork)
2. Ground meat - *untextured*
    a. Red meat burger (beef/pork)
    b. Ground meat (beef/pork)
    c. Meatballs (beef/pork)
3. Breakfast style meats - *routine/context-based*
    a. Sausage (beef/pork)
    b. Bacon (pork)
    c. Breakfast sandwich sausage (beef/pork)
4. Chicken - *separate to red meat*
    a. Unfried chicken
    b. Fried chicken/fried wings (chicken/poultry)
5. Food-based dairy - *separate to meat*
    a. Savory food-based dairy (cheese/cream cheese)
    b. Sweet food-based dairy (plant milk/plant butter)
6. Egg

<u>Total number of coefficients of interest per model and number of models</u>

Coefficients of primary interest with Tier 1 (2 exposures * 6 categories) = 12 models, 12 coefficients

Total coefficients (primary or secondary):
Tier 1: (6 primary outcomes) * (2 exposures) = 12 models
Tier 2: (6 primary outcomes) * (2 exposures) = 12 models

Total: 24 models, 24 coefficients

Analysis set 3 (A3): ITS of MPBA introductions on general ABF sales

Variables

<u>Outcomes (Y_t):</u>

1. We will use the same primary outcomes as in Analysis 1.

<u>Exposure (X_t):</u>

1. The MPBA introduction exposure as an indicator (post vs. pre introduction)

<u>Covariates (C_t):</u>

1. Covariates listed for Analysis 1

We will include within each model individual eligible MPBA and restaurant combinations, including as many time points as possible (within the restaurant) before and after the introduction date such that there were no major menu changes. Multiple introductions within an individual restaurant will be included as separate shocks.

<u>Total number of coefficients of interest per model and number of models</u>

Coefficients of primary interest with Tier 1
(3 outcomes) * (2 parameters - slope and intercept) = 3 models, 6 coefficients

Total coefficients (primary or secondary):
Tier 1: (3 primary outcomes) * (2 parameters - slope and intercept) = 3 models, 6 coefficients
Tier 2: (3 primary outcomes) * (2 parameters - slope and intercept) = 3 models, 6 coefficients

Total: 6 models, 12 coefficients

Analysis set 4 (A4): ITS of MPBA introductions on targeted ABF

Variables

Outcomes (Y_t):

1. We will use the same primary outcomes as in Analysis 2 except that we will remove animal product classes A_i for which there was no introduced MPBA (here, egg)

Exposure (X_t):

1. Exposure listed for Analysis 3

Covariates (C_t):

1. Covariates listed for Analysis 1

Animal product classes (A_i):

1. Whole meat - *like muscle*
    a. Lamb chunked meat (lamb)
    b. Chunked meat (beef/pork)
    c. Pulled pork (pork)
2. Ground meat - *untextured*
    a. Red meat burger (beef/pork)
    b. Ground meat (beef/pork)
    c. Meatballs (beef/pork)
3. Breakfast style meats - *routine/context-based*
    a. Sausage (beef/pork)
    b. Bacon (pork)
    c. Breakfast sandwich sausage (beef/pork)
4. Chicken - *separate to red meat*
    a. Unfried chicken
    b. Fried chicken/fried wings (chicken/poultry)
5. Food-based dairy - *separate to meat*
    a. Savory food-based dairy (cheese/cream cheese)
    b. Sweet food-based dairy (plant milk/plant butter)

Total number of coefficients of interest per model and number of models

Similar to Analysis 3, but for each eligible MPBA, the outcomes will be the targeted animal product outcomes from Analysis 2.

Coefficients of primary interest with Tier 1
(5 outcomes) * (2 parameters - slope and intercept) = 5 model, 10 coefficients

Total coefficients (primary or secondary):
Tier 1: (5 primary outcomes) * (2 parameters - slope and intercept) = 5 models, 10 coefficients

Tier 2: (5 primary outcomes) * (2 parameters - slope and intercept) = 5 models, 10 coefficients

Total: 10 models, 20 coefficients

Analysis set 5 (A5): ITS of MPBA introductions on fixed-customer-base general ABF sales

The goal of this analysis is to account for the possibility that effects seen in main analyses are due to a changing customer base. Stated otherwise, there could be time-invariant unmeasured confounding such that unmeasured person characteristics (e.g., being a vegetarian or vegan) affect whether someone comes to the restaurant during the treated and untreated periods, and also affect what they order.

Customer identifiers within the data
Total number of transactions with customer IDs in all restaurants: 1,505,851
Total number of transactions in all restaurants: 2,066,627
Percentage of transactions with customer IDs (not counting a single restaurant with no customer IDs): **72.87%**

To address this issue, we will fit a fixed-effects model with fixed effects of the customer. For computational feasibility, rather than including dummy variables for every customer in the model, we will mean-center each customer's Y values by their within-person mean. These means will be rounded to a whole number, such that the mean-centered observations can remain as count variables, and fit within the INGARCH paradigm. We will exclude all customers that do not have at least 1 data point pre- and post-MPBA. By including gender data, we will assess effect-measure modification (EMM) by gender.

Outcomes (Y_t):

1. Absolute count of customer (mean-centered within-customer) non-vegan dish purchases
2. Absolute count of customer (mean-centered within-customer) meat dish purchases
3. Absolute count of customer (mean-centered within-customer) chicken and fish dish purchases

Exposures (X_t):

1. Same as Analysis 3
2. Customer gender
3. Interaction of gender with exposure from Analysis 3

Covariates (C_t):

1. Covariates listed for Analysis 1

The restaurants have the current number of unique customer transactions with purchases both within two months before and two months after the promo item. They respectively have gender data in the following percentages within that 4 month period covering the introduced MPBA. The percentages are similar if you look at all customers that had purchases before and after but not necessarily within the 4 month period.

Total number of coefficients of interest per model and number of models

Coefficients of primary interest with Tier 1
(3 outcomes) * (3 parameters - intercept, slope of gender, interaction of gender with pre/post indicator) = 3 models, 6 coefficients

Total coefficients (primary or secondary):
Tier 1: (3 primary outcomes) * (2 parameters - slope and intercept) = 3 models, 6 coefficients
Tier 2: (3 primary outcomes) * (2 parameters - slope and intercept) = 3 models, 6 coefficients

Total: 6 models, 12 coefficients

Analysis set 6 (A6): ITS of MPBA introductions on fixed-customer-base targeted ABF sales

Similar to the analysis for all animal product sales, we want to estimate effects for targeted ABF sales.

Outcomes (Y_t):

1. Number of sales including animal product A_i by customer (mean-centered within customer), aging removing animal product classes for which there was no introduced MPBA

Exposures (X_t):

1. Same as Analysis 5

Covariates (C_t):

1. Covariates listed for Analysis 1

Total number of coefficients of interest per model and number of models

Coefficients of primary interest with Tier 1
(5 outcomes) * (2 parameters - slope and intercept) = 5 models, 10 coefficients

Total coefficients (primary or secondary):
Tier 1: (5 primary outcomes) * (2 parameters - slope and intercept) = 5 models, 10 coefficients
Tier 2: (5 primary outcomes) * (2 parameters - slope and intercept) = 5 models, 10 coefficients

Total: 10 models, 20 coefficients

All Totals for Models

A1
T1
36 (6 outcome x 6 exposures) models
Reasoning: 3 primary outcomes, 3 secondary outcomes
T2
36 (6 outcome x 6 exposures) models
Reasoning: Same as above

A2
T1
12 (6 outcomes x 2 exposures) models
Reasoning: 6 animal products categories (consolidated from the original 14) as outcomes and two (product dependent, restaurant dependent) exposure, the binary and count exposures
T2
12 (6 outcomes x 2 exposures) models
Reasoning: Same as above

A3
T1
3 (outcomes) models
Reasoning: Assuming only the 3 primary outcomes and one exposure, the ITS
T2
3 (outcomes) models
Reasoning: Same as above

A4
T1
5 (outcomes) models
Reasoning: 5 animal products categories (consolidated from the original 14) as outcome and one (product dependent, restaurant dependent) exposure, the ITS
T2
5 (outcomes) models
Reasoning: Same as above

A5
T1
3 (outcomes) models
Reasoning: Only the primary 3 outcomes and one exposure, the ITS
T2
3 (outcomes) models
Reasoning: Same as above

A6
T1
5 (outcomes) models
Reasoning: 5 animal products categories (consolidated from the original 14) as outcomes and one (product dependent, restaurant dependent) exposure, the ITS
T2
5 (outcomes) models
Reasoning: Same as above

Possible Exploratory Analyses

In addition to the main analyses listed above, we may also explore whether restaurant-level characteristics moderate substitution effects on ABF or the lack thereof. This could only be viably done on the encompassing Tier 2 set of restaurants (n=20), as the more stringent Tier 1 set (n=6) would be too small to estimate restaurant-level differences. Analyzing characteristics such as the veg-friendliness, dining style, cuisine type, location of restaurants, and customer characteristics could provide useful information for exploring heterogenous substitution effects (e.g., perhaps MPBAs do have a substitution effect on ABFs, and this effect is twice as large in female-dominated restaurants or restaurants in affluent cities). Being more exploratory in nature, these analyses would ideally provide further interesting insights into the data beyond the analyses targeted at the primary research questions, and ideally motivate further research into more detailed and specific research questions.

Some examples of potential variables that can be analyzed across restaurants are:
- Restaurant veg-predominance, perhaps binary (similar to menu presence analyses, but done at the restaurant level)
- Restaurant gender proportion (similar to fixed-customer-base analyses, but done at the restaurant level)
- Restaurant speciality (dine-in versus take-out)
- Location affluence (city/zip), thresholded at the US median income level
- Location population (city/zip), perhaps threshold 100,000 to distinguish small and large cities
- Location age (city/zip)
- Locational political leaning (city/zip)